Title

Automated Macro-Financial Risk Intelligence Pipeline
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Introduction

The objective of this project was to design and implement an end-to-end automated macro-financial analytics pipeline using Python.

The system retrieves real economic data from external APIs, processes and standardizes multiple macroeconomic indicators, applies a rule-based risk scoring model, and generates automated executive-level reports in Excel and PDF format.

The primary goal was to simulate a Financial Planning and Analysis (FP&A) macroeconomic monitoring system capable of supporting strategic decision-making through data-driven insights and fully automated reporting.

Methodology

The project was structured into modular components covering data acquisition, transformation, analysis, storage, and reporting automation.

Macroeconomic data was retrieved programmatically using the Federal Reserve Economic Data (FRED) API. Four key indicators were selected:

- Consumer Price Index (CPI)
- Unemployment Rate
- Federal Funds Rate
- 10-Year Treasury Yield

These indicators were chosen because they capture complementary dimensions of macro-financial risk, including inflation pressure, labour market conditions, monetary policy stance, and long-term market expectations.

Raw API responses were transformed into structured Pandas DataFrames. Data preparation included:

- Date normalization
- Numeric conversion
- Handling missing values
- Sorting and alignment by time index

To ensure comparability, all indicators were standardized to a monthly frequency. For higher-frequency data such as bond yields, monthly averages were used to reduce noise and improve analytical consistency.

Feature Engineering

To enhance analytical value, several financial features were engineered:

- Month-over-Month (MoM) percentage change
- Year-over-Year (YoY) percentage change
- Rolling averages (3 and 6 months)

These transformations enable the identification of short-term momentum, long-term trends, and potential turning points, which are critical for FP&A scenario analysis.

Analysis and Insight Generation

A rule-based macro risk engine was developed to simulate monitoring logic commonly used in financial planning environments.

Risk signals are generated based on conditions such as:

- Elevated inflation levels
- Rising unemployment trends
- High interest rate environments
- Bond yield stress

Each condition contributes to a cumulative risk score classified as LOW, MEDIUM, or HIGH.

To provide a more interpretable macroeconomic view, a composite **Macro Financial Stress Index (MFSI)** was created using standardized Z-scores across key indicators. This index allows macroeconomic conditions to be mapped into regimes ranging from VERY LOW to CRITICAL.

The model also generates historical context by identifying top stress periods, enabling comparison between current conditions and past macroeconomic environments.

SQL Integration

To simulate an enterprise-level data workflow, the processed dataset is stored in a local SQLite database.

SQL queries are used to generate:

- An executive snapshot of the latest period
- A rolling 12-month trend view

These outputs are automatically exported and included in the Excel report, demonstrating integration between Python-based data processing and SQL-driven business intelligence workflows.

Automation and Reporting

The entire workflow is automated through a centralized pipeline script that orchestrates:

- API data ingestion
- Data transformation and feature engineering
- Risk scoring and regime classification
- SQL storage and query execution
- Automated report generation

The system produces:

- A structured Excel report with analytical outputs
- An executive PDF briefing
- Visual charts illustrating macro trends
- Log files for execution tracking

This automation significantly reduces manual effort while ensuring consistency, reproducibility, and scalability.

Business Impact

The system bridges raw macroeconomic data and structured financial interpretation.

It enables:

- Early detection of inflationary and tightening cycles
- Quantified macro stress monitoring through a composite index
- Comparison with historical stress regimes
- Automated executive reporting without manual intervention

This demonstrates how financial data analytics can directly support budgeting, forecasting, risk management, and capital allocation decisions.

Conclusion

This project demonstrates a complete end-to-end automation solution integrating API-based data acquisition, financial feature engineering, rule-based risk modelling, SQL storage, and automated reporting.

The system goes beyond data processing by transforming macroeconomic information into structured and actionable insights.

Ultimately, the solution illustrates how macroeconomic data can be converted into a scalable and automated decision-support tool, improving both analytical depth and business usability in financial environments.